Exact Spectral Majorants for a Wilson-Dirac Interval

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Abstract

For the negative-energy band of an infinite one-dimensional Wilson–Dirac chain, compressed to a finite interval of consecutive two-orbital cells, we construct an exact longitudinal–transverse decomposition of the difference between the massive and massless covariances. A positive Toeplitz majorant controls the longitudinal component, while the transverse component satisfies a pairing inequality in the eigenbasis of the massless reference. Direct Fourier analysis and finite Jacobi bond identities establish, at every interval length, a critical inverse-linear spectral-tail bound with explicit constants. The resulting estimates apply to the fixed-parameter mass scaling of the relative-entropy problem and establish the positive-majorant and staggered-pairing conditions required in its modular remainder analysis. We also obtain an exact finite-interval Hilbert–Schmidt majorization. The reference-modular approximation and the other asymptotic branches require separate analysis.

1. Covariance and reference geometry

For a dimensionless mass $m \geq 0$, let

$$h_m(k) = \sin k\, \sigma_x + (1 + m - \cos k)\sigma_z, \qquad P_m(k) = \frac{1}{2}\left(I - \frac{h_m(k)}{E_m(k)}\right),$$

$$E_m(k)^2 = m^2 + 4(1+m)\sin^2(k/2), \qquad -\pi < k < \pi.$$

The multiplier P_m is an orthogonal projection for almost every momentum. Its compression to L consecutive cells defines the $2L \times 2L$ covariance $C_{L,m}$. We use $N = 2L$, fix $\mu > 0$, and set

$$m = \frac{\mu}{L}, \qquad r = \frac{m}{m+2} = \frac{\mu}{N+\mu} \in (0,1).$$

The on-site unitaries

$$W = \frac{I + i\sigma_x}{\sqrt{2}}, \qquad D = \mathrm{diag}(-i, 1)$$

followed by cell staggering $(-1)^a$ give an interval-preserving unitary U_L. Fourier integration of the massless symbol gives

$$B_N := U_L C_{L,0} U_L^*, \qquad (B_N)_{ij} = \begin{cases} \frac{1}{2}, & i = j, \\ \dfrac{\sin[\pi(i-j)/2]}{\pi(i-j)}, & i \neq j. \end{cases}$$

Indeed, before staggering, the rotated massless projector has upper-right Bloch entry $i\,\mathrm{sgn}(k)e^{-ik/2}/2$. In the Fourier convention above, the elementary integral

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{sgn}(k) e^{i(n+1/2)k} dk = \frac{i}{\pi(n+1/2)} \qquad (n \in \mathbb{Z})$$

and its companion with $n - 1/2$ give the displayed coefficients after the diagonal phase transformation and staggering. The interval compression of the half-band Fourier projector obeys $0 < B_N < I$.

Define the real Jacobi matrix J_N by its off-diagonal entries

$$(J_N)_{j-1,j} = (J_N)_{j,j-1} = \alpha_j := \frac{j(N-j)}{N}, \qquad 1 \le j < N,$$

with $\alpha_0 = \alpha_N = 0$. An entrywise calculation, using $(d-1)T_{d-1} + (d+1)T_{d+1} = 0$ for $T_d = \sin(\pi d/2)/(\pi d)$, establishes $[B_N, J_N] = 0$. The Jacobi matrix is irreducible, so its spectrum is simple. Hence the two reference matrices admit a common orthonormal eigenbasis.

Throughout, $A_N = U_L C_{L,m} U_L^*$ and $\Delta_N = A_N - B_N$. These are *compressions of the full-line band projections*, not finite-chain open-boundary ground-state projectors.

2. Exact longitudinal–transverse decomposition

For $k \in (-\pi, 0) \cup (0, \pi)$, write

$$x = \sin(k/2), \quad c = \cos(k/2), \quad d_r = \sqrt{x^2 + r^2 c^2}, \quad u_r = \frac{|x|}{d_r}, \quad a_r = \frac{rc}{d_r}, \quad g_r = 1 - u_r.$$

Notice $0 \le g_r \le 1$ and $u_r^2 + a_r^2 = 1$. In the on-site DW-rotated Bloch basis, the two Hermitian involutions

$$Q_0(k) = \begin{pmatrix} 0 & i\,\mathrm{sgn}(k) e^{-ik/2} \\ -i\,\mathrm{sgn}(k) e^{ik/2} & 0 \end{pmatrix}, \qquad Q_\perp(k) = \begin{pmatrix} 0 & e^{-ik/2} \\ e^{ik/2} & 0 \end{pmatrix}$$

satisfy

$$Q_0^2 = Q_\perp^2 = I, \qquad Q_0 Q_\perp + Q_\perp Q_0 = 0.$$

In this basis the occupied-band involution is $Q_m(k) := 2(DW)P_m(k)(DW)^* - I = u_r Q_0 + a_r Q_\perp$. Consequently the difference of full-line projectors has the exact decomposition

$$P_m - P_0 = \mathsf{S} + \mathsf{D}, \qquad \mathsf{S}(k) = -\frac{g_r(k)}{2} Q_0(k), \quad \mathsf{D}(k) = \frac{a_r(k)}{2} Q_\perp(k), \quad \mathsf{G}(k) = \frac{g_r(k)}{2} I. \tag{2.1}$$

Here the full-line objects are understood after the on-site rotation; the cell-staggering conjugation can be applied to all of them simultaneously. Fibrewise,

$$-\mathsf{G} \le \mathsf{S} \le \mathsf{G}, \quad [\mathsf{G}, Q_0] = 0, \quad \{\mathsf{D}, Q_0\} = 0, \tag{2.2}$$

$$\mathsf{D}^2 = \mathsf{G} - \mathsf{G}^2 \le \mathsf{G}, \qquad \boxed{(P_m - P_0)^2 = \mathsf{G}}. \tag{2.3}$$

All identities hold as full-line Fourier multipliers; exceptional momentum points of measure zero do not affect the operators. Denote the interval compressions, including staggering, by S_N, D_N, G_N, respectively. Then $\Delta_N = S_N + D_N$ exactly.

Theorem 1 (positive majorant and exact trace). For every $L \ge 1$ and $\mu > 0$,

$$\boxed{-G_N \le S_N \le G_N, \qquad G_N \ge 0, \qquad \mathrm{Tr}\, G_N = \frac{L}{\pi}\left(\pi - \frac{2\arccos r}{\sqrt{1-r^2}}\right) \le \frac{\mu}{\pi}.} \tag{2.4}$$

Proof. Compression preserves Loewner order in (2.2). Since G is a scalar two-orbital Fourier multiplier,

$$\operatorname{Tr} G_N = \frac{L}{2\pi}\int_{-\pi}^{\pi} g_r(k)\,dk = \frac{L}{\pi}\int_0^{\pi} g_r(k)\,dk.$$

With $q = k/2$, substitution $z = \cos q$ gives

$$\int_0^{\pi} g_r(k)\,dk = \pi - 2\int_0^1 \frac{dz}{\sqrt{1-(1-r^2)z^2}} = \pi - \frac{2\arccos r}{\sqrt{1-r^2}}.$$

Let $M(r)$ denote this last expression, with $M(0) = 0$. Its derivative is

$$M'(r) = \frac{2}{1-r^2} - \frac{2r\arccos r}{(1-r^2)^{3/2}} \le 2,$$

because $\arccos r \ge r\sqrt{1-r^2}$, equivalently $\theta \ge \sin\theta\cos\theta$ for $\theta = \arccos r \in [0,\pi/2]$. Thus $M(r) \le 2r$, and $2Lr = \mu N/(N+\mu) \le \mu$. □

Theorem 2 (staggered pairing). Let v be any normalized vector supported in the selected interval. Set $b = \langle v, B_N v\rangle$, $q = \langle v, G_N v\rangle$. Then

$$\boxed{|\langle v, D_N v\rangle| \le 2\sqrt{q\,b(1-b)}.} \tag{2.5}$$

In particular this holds for every common normalized reference eigenvector of B_N and J_N.

Proof. Work before compression with the full-line massless projector P_0, transverse operator D, and commuting positive operator G. Equation (2.2) implies $P_0\mathsf{D}P_0 = (I-P_0)\mathsf{D}(I-P_0) = 0$. Write $p = P_0 v$, $z = (I-P_0)v$, and let

$$X = \langle p, \mathsf{G}p\rangle, \qquad Y = \langle z, \mathsf{G}z\rangle.$$

Because $[\mathsf{G}, P_0] = 0$, we have $X + Y = q$, while $\|p\|^2 = b$ and $\|z\|^2 = 1-b$. Moreover,

$$|\langle v, \mathsf{D}v\rangle| = 2|\operatorname{Re}\langle p, \mathsf{D}z\rangle| \le 2\min\{\sqrt{X(1-b)}, \sqrt{Yb}\},$$

by Cauchy–Schwarz and $\mathsf{D}^2 \le \mathsf{G}$. For nonnegative X, Y with $X + Y = q$, $\min\{X(1-b), Yb\} \le b(1-b)q$: if $X \le bq$, use the first term; otherwise $Y < (1-b)q$, and use the second. Since v is interval-supported, its full-line expectation equals $\langle v, D_N v\rangle$. □

3. A uniform critical Jacobi spectral tail

Let (λ_j, v_j) be the eigenpairs of J_N, with $\|v_j\| = 1$, and define the positive measure

$$\nu_N := \sum_{j=1}^{N} \langle v_j, G_N v_j\rangle\, \delta_{|\lambda_j|}.$$

Theorem 3 (explicit critical tail). For every even $N = 2L \ge 2$, every $\mu > 0$, and every $t > 0$,

$$\boxed{\nu_N((t,\infty)) \le \min\left\{\frac{\mu}{\pi}, \sqrt{\frac{2}{15}}\frac{\mu^2}{\pi t} + \frac{\mu}{3\pi t^2}\right\}.} \tag{3.1}$$

Proof. The first estimate is Theorem 1. To obtain the second, express the same-parity blocks of G_N as Fourier mixtures. For $\varepsilon \in \{0,1\}$ define the length-N vector

$$\phi_{k,\varepsilon}(2a+\varepsilon) = (-1)^a e^{ika}, \quad \phi_{k,\varepsilon}(2a+1-\varepsilon) = 0, \qquad 0 \le a < L.$$

The positive Toeplitz representation is

$$G_N = \frac{1}{2}\sum_{\varepsilon=0}^{1} \int_{-\pi}^{\pi} g_r(k) |\phi_{k,\varepsilon}\rangle\langle\phi_{k,\varepsilon}| \, \frac{dk}{2\pi}, \qquad \|\phi_{k,\varepsilon}\|^2 = L. \tag{3.2}$$

For either parity, application of J_N yields adjacent-bond differences of the form $\alpha_j - e^{ik}\alpha_{j+1}$, up to phases. Reflection about the centre of the interval interchanges the two parities, making their squared-norm sums identical. Direct polynomial summation gives

$$\frac{1}{L}\sum_{j \equiv \varepsilon\,(2)} (\alpha_j - \alpha_{j+1})^2 = \frac{N^2-1}{3N^2},$$

$$\frac{4}{L}\sum_{j \equiv \varepsilon\,(2)} \alpha_j \alpha_{j+1} = \frac{2(N^2-1)(N^2-4)}{15N^2},$$

where the sums run over $0 \le j < N$, and $\alpha_0 = \alpha_N = 0$. Consequently

$$\frac{\|J_N \phi_{k,\varepsilon}\|^2}{L} = \frac{N^2-1}{3N^2} + \frac{2(N^2-1)(N^2-4)}{15N^2}\sin^2(k/2) \le \frac{1}{3} + \frac{N^2 k^2}{30}. \tag{3.3}$$

For $\Pi_t = \mathbf{1}_{\{|J_N| > t\}}$, the spectral theorem implies

$$\|\Pi_t \phi\|^2 \le \min\{\|\phi\|^2, t^{-2}\|J_N\phi\|^2\}.$$

Substituting (3.3) into (3.2) and using the evenness of g_r,

$$\nu_N((t,\infty)) \le \frac{L}{\pi}\int_0^{\pi} g_r(k) \min\left\{1, \frac{1}{3t^2} + \frac{N^2k^2}{30t^2}\right\} dk. \tag{3.4}$$

The elementary scalar bounds

$$\int_0^{\pi} g_r(k)\, dk \le 2r, \qquad g_r(k) = 1 - \left[1 + r^2\cot^2(k/2)\right]^{-1/2} \le \frac{2r^2}{k^2} \quad (0 < k \le \pi) \tag{3.5}$$

follow from Theorem 1, $1-(1+x)^{-1/2} \le x/2$, and $\tan(k/2) \ge k/2$. Set $k_* := \sqrt{30}\,t/N$. Bound the integrand in (3.4) by the second argument on $[0, \min(k_*, \pi)]$, and by 1 on $[k_*, \pi]$ when the latter interval is nonempty. Equation (3.5) yields, whether or not $k_* > \pi$,

$$\begin{aligned}\nu_N((t,\infty)) &\le \frac{L}{\pi}\left[\frac{2r}{3t^2} + \frac{N^2}{30t^2}(2r^2 k_*) + \frac{2r^2}{k_*}\right] \\ &= \frac{L}{\pi}\left[\frac{2r}{3t^2} + \frac{4Nr^2}{\sqrt{30}\,t}\right].\end{aligned}$$

Finally $2Lr = Nr \le \mu$ and $4LNr^2 = 2N^2r^2 \le 2\mu^2$. Since $2/\sqrt{30} = \sqrt{2/15}$, this is exactly the second estimate in (3.1). □

4. A finite compression inequality and continuum consequence

Corollary 4 (covariance-distance domination). At every L and $\mu > 0$,

$$\boxed{0 \le \Delta_N^2 \le G_N, \qquad \|A_N - B_N\|_{\mathrm{HS}}^2 \le \operatorname{Tr} G_N \le \frac{\mu}{\pi}.} \tag{4.1}$$

Proof. If χ is the interval projection and $T = P_m - P_0$ after the same full-line unitary, then $T = T^*$ and $T^2 = \mathsf{G}$ by (2.3). The identity

$$\chi T^2 \chi - (\chi T \chi)^2 = \chi T (I - \chi) T \chi \ge 0$$

proves the operator inequality; take the finite trace and use Theorem 1. □

For μ fixed and $L \to \infty$, the exact trace in (2.4) tends to μ/π, since g_r has integrated mass $2r + O(r^2)$. Thus whenever the finite covariance differences converge in Hilbert–Schmidt norm to the continuum kernel of *Hilbert-Schmidt Scaling Limit for Wilson-Dirac Covariances*, with squared norm $H(\mu)$, (4.1) independently yields

$$0 \le H(\mu) \le \frac{\mu}{\pi}.$$

There is an independent continuum verification. For the rescaled Bloch-projector difference

$$F_\mu(t) = -\frac{1}{2}\left[\left(\frac{t}{\sqrt{t^2+\mu^2}} - \operatorname{sgn} t\right)\sigma_x + \frac{\mu}{\sqrt{t^2+\mu^2}}\sigma_z\right],$$

one calculates $\|F_\mu(t)\|_F^2 = 1 - |t|/\sqrt{t^2+\mu^2}$ and therefore, by direct integration and Plancherel,

$$\int_{\mathbb{R}} \|\widehat{F}_\mu(s)\|_F^2 ds = \frac{1}{2\pi}\int_{\mathbb{R}} \|F_\mu(t)\|_F^2 dt = \frac{\mu}{\pi}.$$

The interval overlap formula for $H(\mu)$ has weight $0 \le (1-|s|)_+ \le 1$, so the same bound follows without reference to finite-size eigenvalues.

The finite identity also identifies the trace deficit as cross-interval leakage:

$$\operatorname{Tr} G_N - \|\Delta_N\|_{\mathrm{HS}}^2 = \|(I-\chi)T\chi\|_{\mathrm{HS}}^2.$$

5. Consequences for the relative-entropy problem

Theorems 1–3 establish for the specified Wilson chain the Fourier splitting, positive-majorant trace bound, transverse reference-mode pairing, and critical Jacobi spectral tail. The proofs require no large-L hypothesis and remain valid for arbitrary positive μ at each finite L.

In the relative-entropy decomposition, the modular term involves $E_N = \log(I - B_N) - \log B_N + N\eta(J_N/N)$. The "Positive weighted modular control" theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*, section "Global envelope and weighted consequences," supplies a bounded longitudinal term and the variance estimate $O((\log N)/N^2)$. The entropy trace-path limit is established in *Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States*, equation (B9). For the specified compressed infinite-chain Wilson model at fixed $\mu > 0$, the additional convergence estimates and full constant-term expansion are established in *Finite Part of Wilson–Dirac Relative Entropy*, Theorem 1 and its proof, and sharpened in *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*, Theorem 3, Corollary 4 and Theorem 6. Their exact concentration-connection and boundary inputs remain those stated in those papers. The present paper supplies the projector, pairing, and positive-tail inputs.

References

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